

Run a Microwave Heating Cycle SOP (1x Cycle, in situ)

One episode runs one heating cycle at the microwave where it lives. The base is **passive**: it has no drive of its own, so it is pushed by hand to the counter in front of the microwave and locked there, and nothing is carried away to a table. Everything the episode touches is already there when recording starts: the microwave in its cabinet run, and the plate, the food tub, the spoon, and the cover on the counter to the right of it.

The episode runs these nine actions in this order and no other: **plate the food, cover it, open the door, center the plate, set the time, run the cycle, retrieve the plate hot-safe, uncover it, stir it.**

The microwave is worked **as found**. The door is shut at the start, the cavity is empty, the plate is empty, and the food tub is open. The episode ends with the food plated, heated, uncovered, and stirred, standing on the counter where the plate started.

This is an in-situ task, and three things follow from that. First, there is a **cabinet directly above the microwave**, and the cavity has a ceiling of its own, so **every approach is from the front, straight in, level**. No gripper comes down into the cavity from above. Second, the **microwave is never leaned on and never pushed** — no gripper, wrist, or forearm rests on the door, the door mouth, or the control panel, and no press is ever hard enough to shift the appliance. Third, the **door stands open at its stop on its own**, so neither arm holds it and both arms stay free for the plate.

The plate is carried by both grippers, one on each side of its rim. It goes in that way and it comes out that way. Both grippers have **heat pads** fitted to their fingers before the episode starts, so the same grip works on a cold plate and on a hot one.

After the cycle everything is hot. From the moment the cycle ends, the grippers touch only the **plate rim**, the **cover tab**, and the **spoon handle**. Nothing touches the food, the cover dome, the turntable, or the inside of the cavity.

Arm roles are fixed and never swap. The **right gripper is the working gripper**: the spoon, the cover, the door handle, the control panel, the stir, and the right side of the plate rim. The **left gripper is the holder**: it holds the food tub still while the food is scooped, holds the plate still while the cover goes on and comes off and while the food is stirred, and takes the left side of the plate rim in the two carries.

Setup

Complete the base positioning and both checklists before starting an episode.

Cell configuration

- **Environment camera:** 900 mm.
- **cell_type:** bimanual

Base positioning

The base is **passive**. It has no drive of its own, it is pushed into place by hand, and it is never steered, nudged, or repositioned once recording starts. It is parked once, before recording, and does not move again until the episode is over.

1. Push the base by hand up to the counter and stop it **square to the front of the microwave**, so the front of the appliance runs straight across the frame and neither end of it sits nearer than the other.
2. Stop it **centered on the door mouth**, so the left side and the right side of the mouth are the same distance out from the middle of the base.
3. Stop it **close enough** that both grippers reach the back of the turntable straight in through the door mouth without either arm extending, and **far enough** that neither arm, wrist, nor any part of the base touches the microwave, the open door, or the cabinet above while both arms work.
4. Check the **height band**: with the base parked, both grippers enter the door mouth level, carry the plate from the counter to the back of the turntable, and the **right gripper** reaches every key on the control panel, all without a wrist or forearm fouling the cabinet above.
5. Check the **door swing**: the **right gripper** reaches the door handle with the door shut and can carry it through its whole swing to its stop without extending, and the open door is clear of both arms' path to the door mouth.
6. Check the **counter**: the **right gripper** reaches into the food tub, reaches the spoon, and reaches the cover, and both grippers reach the plate's rim where it stands, all without extending and without knocking anything.
7. Lock or brake the base. Push it firmly once by hand: it must not roll, creep, or turn.
8. If any of lines 1 to 6 fails, push the base to a new park by hand and start again at line 1. Do not work a microwave the arms cannot reach comfortably.

The base stays locked and still for the whole episode. Nothing moves it: no arm leans on the cabinet run hard enough to shift it, nothing touches it by hand, and it is never repositioned mid-task. A base that moves after recording starts ends the episode.

Hardware checklist

1. Cameras are on and recording.
2. The environment camera is centered on the door mouth and its frame includes the whole turntable, the door shut and open, the control panel, and the plate, food tub, spoon, and cover on the counter.

3. The camera reads the turntable from the front, so whether the plate sits centered and whether it clears the cavity walls are readable.
4. The camera reads the plate from the front, so how much food is on it, whether the cover is seated, and whether the food has been stirred are readable.
5. The camera reads the control panel, so which keys are pressed and what the display shows are readable.
6. The camera reads the door front once it is shut, so a door standing ajar is readable.
7. Both arms are at home with grippers open.
8. **Both grippers have their heat pads fitted** to their fingers, seated flat and not loose. They are fitted before recording starts and are never taken off or put on during the episode.
9. The **right arm** reaches the food tub, the spoon, the cover, the door handle, every key on the control panel, the plate's right rim, and the back of the turntable, without extending to a joint limit.
10. The **left arm** reaches the food tub, the plate's left rim, and the back of the turntable, without extending. It reaches no key on the control panel and not the door handle.
11. Both grippers enter and leave the door mouth **from the front and level**. Neither comes down into the cavity from above, and neither wrist nor forearm touches the cabinet above on the way in or out.
12. The two arms do not collide while they carry the plate together, and the open door is clear of both arms' path to the door mouth and to the control panel.
13. If a place cannot be reached, re-park the base by the Base positioning steps until lines 9 to 12 hold.

Materials checklist

1. The **microwave** stands where it lives, in its cabinet run. It is not moved, not leaned on, and not pushed at any point.
2. There is a **cabinet directly above the microwave**, so the only way in is the front-facing door mouth and never the top.
3. The **door** is hinged on the left, with its handle on the right. It swings open to the left and **stands at its stop on its own**, without being held and without drifting shut.
4. The door shuts with a latch that holds it closed. A correctly shut door meets the frame all round, stands flush with the front of the appliance, and does not spring back open.
5. The microwave will not run with the door open, and its light and turntable stop the moment the door is opened.
6. The **control panel** sits to the right of the door mouth. It carries number keys and a **Start** key, all readable from the front, and it beeps when the cycle ends.
7. The **turntable** is in place, sits level, and turns freely when the cavity is empty.

8. The **plate** stands on the counter to the right of the microwave, empty, dry, and flat. It does not rock.
9. The plate has a **broad flat rim** and a flat **inner circle**, so where the food goes is never a judgment call and both grippers have somewhere to take hold.
10. The plate is small enough to sit on the turntable with clear space all round it, so it turns without catching a cavity wall, and low enough to clear the cavity ceiling with the cover on.
11. The plate is heat-proof and stays whole through a cycle, and its rim can be gripped by a padded gripper when it comes out hot.
12. The **cover** stands on the counter to the right of the plate, dome up, with its **cover tab** on top. It is vented, so steam leaves it while the cycle runs.
13. The cover sits down on the plate's rim and **leaves the outer edge of the rim clear**, so both grippers can take the plate with the cover on.
14. The **food tub** stands open on the counter to the right of the cover. It does not slide when the spoon is drawn through it.
15. The tub holds enough food for **three level scoops** with food left over, and the food is thick enough to stay on the spoon and to hold a mound on the plate.
16. The **spoon** lies flat on the counter beside the food tub, bowl up, with its handle toward the front. Its handle stays cool and is long enough to stir with the bowl down on the plate.
17. The counter between the plate, the cover, the food tub, and the spoon is clear, so an arm carrying the spoon, the cover, or the plate passes over nothing.
18. The counter is deep enough that the covered plate stands on it clear of the front edge, and it is heat-proof where the plate is set down.
19. Nothing else stands on the counter within either arm's reach.
20. The cavity is empty at the start, and the turntable is clean and dry.

Workspace layout

Nothing anywhere is marked or taped out. Every place below is judged by eye against the appliance and the cabinet run themselves — the door mouth, the turntable, and the counter.

- **Microwave:** the appliance the base is parked at. It is never moved, never leaned on, and never pushed.
- **Cabinet above:** the cabinet directly over the microwave. It is what makes every approach a front approach. Nothing ever touches it.
- **Door mouth:** the open front of the cavity. Both grippers go in and come out through it, level and from the front.
- **Cavity:** the inside of the microwave, holding the turntable. Nothing goes into it but the plate and the grippers that carry it.
- **Turntable:** the round plate in the cavity floor. The plate is centered on it.
- **Control panel:** the keys to the right of the door mouth. **Right gripper only.**

- **Door:** the microwave door. Opened, shut, and pressed home by the **right gripper**, and held by neither arm.
- **Plate:** stands on the counter to the right of the microwave at the start and at the end. Carried by both grippers, one on each side of its rim.
- **Cover:** stands on the counter to the right of the plate, dome up. **Right gripper only.**
- **Food tub:** the open tub of food on the counter. **Right gripper** scoops from it, **left gripper** holds it still.
- **Spoon:** lies flat on the counter beside the food tub, bowl up. **Right gripper only.**

Arm assignments

- **Right gripper — the working gripper.** Works the spoon, the cover, the door handle, the control panel, and the stir, presses the door home, and takes the right side of the plate's rim in both carries.
- **Left gripper — the holder.** Holds the food tub still in Step 1, holds the plate still in Steps 2, 8, and 9, and takes the left side of the plate's rim in both carries. It never touches the spoon, the cover, the door, or any key on the control panel.
- Nothing is handed between grippers, and only one thing is moved at a time.

Vocabulary

- **Front approach:** the gripper enters and leaves the door mouth level and from the front, and never comes down into the cavity from above. Every action at the microwave is a front approach.
- **Heat pads:** the pads fitted to both grippers' fingers, so a gripper can take a hot plate rim. They are part of the station hardware, fitted before recording and never touched during the episode. **Unvalidated.**
- **Rim grip:** both grippers close on the plate's outer rim, one on the left side and one on the right side, straight across from each other, with the pads flat on the rim. It is the only way the plate is ever taken hold of.
- **Cover tab:** the tab on top of the cover. It is the only part of the cover a gripper ever touches.
- **Inner circle:** the flat middle of the plate, inside its rim. All the food goes there.
- **Level scoop:** the spoon is drawn through the food and comes up with its bowl full and its load flat with the bowl's edge, with nothing heaped above it and nothing hanging off it.
- **Seated:** the cover stands down on the plate's rim all the way round, with no gap under its rim, and it does not rock when the **right gripper** comes off its tab.
- **Centered on the turntable:** the plate sits on the turntable with the gap between the plate rim and the cavity wall looking the same on the left as on the right, and the plate's front edge inside the door mouth, so nothing catches a wall when the turntable turns.

- **Latched:** the door meets the frame all round, sits flush with the front of the appliance, and does not spring back when the **right gripper** comes off it.
- **Cycle ended:** the microwave has beeped, the display reads zero, the cavity light is out, and the turntable has stopped.
- **Hot parts:** after the cycle, the food, the cover dome, the turntable, and the inside of the cavity. No gripper touches any of them. The plate rim, the cover tab, and the spoon handle are the only things touched from then on.
- **Clear of the front:** the arm is drawn back so that no part of it is over the counter, the control panel, or the door mouth.
- **Stands on its own:** after the grippers move clear, the plate stays flat and does not rock, slide, or tip.
- **Stirred through:** the spoon has been turned through the whole mound, so the food that was at the plate's edge and the food that was in the middle have changed places and the mound reads as one even layer from the front.

Steps

Step 1: Plate the food

Goal: three level scoops of food sit on the plate's inner circle as one low mound, and the spoon is back on the counter.

1.1 Steady the food tub

- With the **left gripper** closed, set it against the left side of the food tub and hold the tub still.
- Hold the tub where it already stands. Do not lift it, do not drag it, and do not tip it.
- The **left gripper** stays there until Step 1.4.

Check: the tub stands where it started and does not move when the **left gripper** settles on it.

1.2 Take the spoon

- With the **right gripper**, close on the spoon handle, not on its bowl.
- Lift the spoon straight up off the counter and carry it level to the tub.

Check: the **right gripper** holds the spoon by its handle, and the bowl hangs clear of the counter.

1.3 Put three level scoops on the plate

- With the **right gripper**, draw the spoon through the food and lift a **level scoop** straight up out of the tub.
- Carry it level to the plate. Do not carry it over the cover or over the control panel.
- Lower the spoon into the plate's inner circle and tip the food off in the middle of the plate.
- Draw the spoon straight up clear of the plate.

- Do this three times. Lay the second and third scoops onto the first, so the food builds into one low mound.
- Keep every scoop inside the inner circle. Do not let food land on the plate rim or on the counter.

Check: three scoops are on the plate, the food sits as one low mound inside the inner circle, and the rim and the counter are clean. If food landed on the rim, push it into the inner circle with the spoon in the **right gripper**.

1.4 Put the spoon back

- With the **right gripper**, carry the spoon back to the counter and lay it flat where it started, bowl up, handle toward the front.
- Open the **right gripper** and draw it clear of the spoon.
- Open the **left gripper** and draw it back off the food tub.

Check: the spoon lies flat on the counter, bowl up, and the food tub stands where it started.

Expected state: the plate holds one low mound of food, the spoon and the cover are on the counter, and the microwave door is still shut.

Step 2: Cover the plate

Goal: the cover is seated on the plate's rim all round, with the outer edge of the rim left clear.

2.1 Hold the plate still

- With the **left gripper** closed, set it against the left side of the plate's rim and hold the plate still.
- Do not lift the plate and do not drag it.
- The **left gripper** stays there until Step 2.3.

Check: the plate stands still on the counter and does not slide when the **left gripper** settles on it.

2.2 Take the cover and set it down over the plate

- With the **right gripper**, close on the **cover tab** on top of the cover. Never close on the dome, the rim, or a vent.
- Lift the cover straight up off the counter and carry it level to the plate.
- Hold it level over the plate, then lower it straight down until its rim stands on the plate's rim all the way round.
- Lower it straight down only. Do not drag it across the plate, and do not tip it on at an angle.
- Open the **right gripper** and draw it straight up off the cover tab.

Check: the cover is **seated** — it stands on the plate's rim all the way round, no gap shows under its rim, it does not rock when the **right gripper** comes off, and the outer edge of the plate's rim is still clear on both sides. If it rocks or sits over the outer edge, take it back up by its tab with the **right gripper** and set it down again.

2.3 Let go of the plate

- Open the **left gripper** and draw it back off the plate's rim.

Check: the covered plate stands on its own on the counter.

Expected state: the covered plate stands on the counter, the spoon and the food tub are where they started, and the door is still shut.

Step 3: Open the door

Goal: the door stands open at its stop, the whole door mouth is clear, and neither arm is holding it.

3.1 Swing the door open

- With the **right gripper**, close on the door handle at about its middle height.
- Draw the door straight out toward the front until the latch releases, then swing it round to the left until it meets its stop.
- Do not jerk it, do not swing it fast, and do not let it hit the cabinet run.
- Open the **right gripper** and draw it back clear of the door.

Check: the door stands at its stop and stays there on its own, the whole door mouth is clear, and the turntable is visible front to back. If the door did not run to its stop, take the handle again with the **right gripper** and swing it the rest of the way. If it drifts back toward the mouth, do not hold it — leave it for the station.

Expected state: the door stands open on its own, the cavity is empty, and the covered plate still stands on the counter.

Step 4: Center the plate on the turntable

Goal: the covered plate sits centered on the turntable, with clear space all round it.

4.1 Take the plate with both grippers

- Bring both grippers to the plate's outer rim: the **left gripper** onto the left side of the rim, the **right gripper** onto the right side, straight across from each other.
- Close both grippers on the rim at the same time, with the pads flat on it.
- Do not close on the cover, do not close on the inner circle, and do not take the plate with one gripper alone.

Check: both grippers hold the plate's rim straight across from each other, and the cover has not shifted.

4.2 Carry the plate in

- Lift the plate straight up off the counter, only high enough to clear it.
- Carry it level to the door mouth and take it straight in from the front. Keep it level the whole way.
- Do not tilt it, do not swing it, and do not let it turn between the grippers.
- Lower it straight down onto the middle of the turntable and set it flat.

Check: the plate stands flat on the turntable, the cover is still seated, and the food is still one mound.

4.3 Center it and come out

- Open both grippers and, with the **right gripper** closed, nudge the plate's rim until it sits **centered on the turntable**.
- Nudge it a little at a time. Do not knock it, and do not push it hard enough to move the microwave.
- Draw both grippers straight out of the door mouth, level.

Check: the gap between the plate rim and the cavity wall looks the same on the left as on the right, no part of the plate or the cover touches a wall, and the plate's front edge is inside the door mouth. If the plate sits near a wall, nudge it back to the middle with the closed **right gripper**.

Expected state: the covered plate stands centered on the turntable, the counter holds the spoon, the food tub, and nothing else, and the door stands open.

Step 5: Set the time

Goal: the door is latched and the display reads thirty seconds.

5.1 Shut the door

- Check first that both grippers are out of the door mouth and empty.
- With the **right gripper**, close on the door handle and swing the door round in front of the mouth in one smooth move. Do not swing it fast, and do not let it slam.
- Open the **right gripper**, then press the door's front face flat, at the handle side, straight back with the gripper closed until the door is **latched**.
- Press straight back only, and do not press hard enough to move the microwave.
- Draw the **right gripper** straight back off the door.

Check: the door is latched — it meets the frame all round, sits flush, and does not spring back when the **right gripper** comes off. If it springs back, press it home once more with the **right**

gripper.

5.2 Press the time in

- With the **right gripper** closed, come straight in to the control panel and press **3**, then **0**, one key at a time.
- Press each key straight in and level, then draw the gripper back off the panel before the next key.
- Do not rest the gripper, wrist, or forearm on the panel, and do not press two keys at once.

Check: the display reads thirty seconds. If it reads anything else, clear it with the **right gripper** and press **3** and **0** again.

Expected state: the door is latched, the display reads thirty seconds, and the cycle has not started.

Step 6: Run the cycle

Goal: the cycle runs to its end with both arms clear and still.

6.1 Start the cycle

- With the **right gripper** closed, press **Start** straight in and level.
- Draw the **right gripper** straight back off the panel.

Check: the cavity light is on and the turntable is turning.

6.2 Stand clear while it runs

- Draw both arms back **clear of the front** and hold them still.
- Touch nothing until the cycle has ended: not the door, not the panel, not the food tub, the spoon, or the cover.
- Wait for the **cycle ended** signs: the beep, the display at zero, the light out, and the turntable stopped.

Check: both arms stayed clear and still, and nothing on the counter moved while the cycle ran.

Expected state: the cycle has ended, the door is shut, and the covered plate is hot on the turntable.

Step 7: Retrieve the plate hot-safe

Goal: the covered plate stands back on the counter where it started, and no gripper touched a hot part.

From here to the end of the episode the grippers touch only the plate rim, the cover tab, and the spoon handle. Nothing touches the food, the cover dome, the turntable, or the inside of the cavity.

7.1 Let it stand, then open the door

- After the cycle has ended, leave everything alone for 5 seconds so the steam settles under the cover.
- With the **right gripper**, close on the door handle and swing the door round to the left to its stop, in one smooth move.
- Open the **right gripper** and draw it back clear of the door.

Check: the door stands at its stop on its own, the plate is still centered on the turntable, and the cover is still seated. If the cover has shifted, leave it — it comes off in Step 8.

7.2 Take the hot plate

- Bring both grippers straight in through the door mouth, level, to the plate's outer rim: the **left gripper** onto the left side, the **right gripper** onto the right side, straight across from each other.
- Close both grippers on the rim at the same time, with the pads flat on it.
- Take hold of the rim only. Do not touch the cover, the food, the turntable, or a cavity wall.

Check: both grippers hold the plate's rim straight across from each other, and nothing else in the cavity has been touched.

7.3 Carry it out and set it down

- Lift the plate straight up off the turntable, only high enough to clear it.
- Draw it straight out through the door mouth, level and slowly. Do not tilt it, do not swing it, and do not let it turn between the grippers.
- Carry it level to the counter and lower it straight down onto the spot the plate started from.
- Set it down flat, then open both grippers and draw them back off the rim.

Check: the plate **stands on its own** on the counter, the cover is still on it, the food has not slid, and nothing has come over the plate rim. If the plate rocks, take the rim again with both grippers and set it down flat.

Expected state: the covered hot plate stands on the counter where it started, the cavity is empty, and the door stands open.

Step 8: Uncover the plate

Goal: the cover stands back on the counter, dome up, and the plate is open with the food still one mound.

8.1 Hold the plate still

- With the **left gripper** closed, set it against the left side of the plate's rim and hold the plate still.
- Do not lift the plate and do not drag it.

- The **left gripper** stays there until Step 9.3.

Check: the plate stands still on the counter and does not slide when the **left gripper** settles on it.

8.2 Lift the cover off

- With the **right gripper**, close on the **cover tab**. Touch nothing else on the cover.
- Lift the cover straight up off the plate, high enough that its rim clears the plate rim.
- As it comes up, tilt its far edge down and away from the front, so the steam under it goes back toward the microwave and not toward the front edge.
- Do not drag the cover across the plate, and do not knock the plate or the food with its rim.

Check: the cover is clear of the plate, the food is still one mound, and nothing has been pulled off the plate with the cover.

8.3 Put the cover back on the counter

- With the **right gripper**, carry the cover level to the counter and set it down where it started, dome up, tab on top.
- Set it down on its rim, then open the **right gripper** and draw it straight up off the tab.

Check: the cover stands on the counter dome up where it started, and it stands on its own without rocking.

Expected state: the plate is open on the counter, the food is one mound, and the cover is back where it started.

Step 9: Stir the food

Goal: the food is stirred through and the spoon is back on the counter.

9.1 Take the spoon

- With the **right gripper**, close on the spoon handle, not on its bowl.
- Lift the spoon straight up off the counter and carry it level to the plate.

Check: the **right gripper** holds the spoon by its handle, and the **left gripper** still holds the plate still.

9.2 Stir the food

- With the **right gripper**, lower the spoon into the mound until its bowl is down against the plate.
- Turn the spoon round through the food **three full turns**, keeping the bowl down against the plate the whole time.
- Keep the spoon inside the inner circle. Do not push food over the plate rim, and do not lift the spoon out between turns.

- Draw the spoon straight up clear of the plate.

Check: the food is **stirred through** and reads as one even layer from the front, and nothing has gone over the plate rim onto the counter. If the mound is still uneven, lower the spoon with the **right gripper** and turn it through once more.

9.3 Put the spoon back

- With the **right gripper**, carry the spoon back to the counter and lay it flat where it started, bowl up, handle toward the front.
- Open the **right gripper** and draw it clear of the spoon.
- Open the **left gripper** and draw it back off the plate's rim.

Check: the spoon lies flat on the counter, bowl up, and the plate stands still where it was set down.

Expected state: the stirred plate stands open on the counter, the cover and the spoon are back where they started, and the cavity is empty.

Step 10: End the episode

- Confirm the plate: it stands on the counter where it started, open, with the food stirred through inside the inner circle.
- Confirm the microwave: the cavity is empty and the turntable is clear.
- Confirm the counter: the cover, the spoon, and the food tub all stand where they started.
- Confirm nothing has been spilled on the counter or the floor.
- Confirm the base has not moved and the microwave has not shifted.
- Correct any failed check before ending.
- Return both arms home with grippers open, then stop recording.

After the episode: reset the workspace

This reset is not recorded.

1. Push the microwave door shut by hand and check the latch holds.
2. Tip the food back into the food tub, then wash and dry the plate and stand it back on the counter to the right of the microwave, empty and flat.
3. Wash and dry the spoon and lay it flat on the counter beside the food tub, bowl up, handle toward the front.
4. Wash and dry the cover and stand it on the counter to the right of the plate, dome up, tab on top.
5. Wipe up any spill on the counter, the turntable, or the cavity floor, and dry it.
6. Check the food tub still holds enough food for three level scoops with food left over, and top it up if it does not.

7. Replace food that has dried out, gone runny, or will no longer hold a mound on the plate.
8. Replace a cover whose tab has come loose, whose rim no longer stands flat on the plate rim, or whose vents are blocked.
9. Replace a plate that has cracked, that rocks on the counter, or whose rim a padded gripper can no longer hold.
10. Check both grippers' heat pads are still seated flat and not loose, and refit or replace any that have shifted.
11. Check the door still latches and does not spring back when it is pushed shut by hand, and that it still stands open at its stop on its own.
12. Check the turntable is seated and turns freely.
13. Check the display is clear and reads nothing before the next episode.
14. Check the base is still locked and parked square, then run the Base positioning steps and both Setup checklists again.

SOP violations

These are actions that break the SOP and are reviewed side by side in the review tool.

How to record a violation in review

For each violation, record the **start timestamp**, **violation name**, and **SOP rule broken**. The visible cue is what the reviewer sees. The coaching note is for retraining and is not an annotation label.

Episode handling

Tag every violation with its timestamp and name. Keep the episode with the violation tag. Do not delete it just because a rule was broken.

Violations

Violation: Base moved during the episode

- **Visible cue:** the microwave shifts in frame, the door mouth changes angle or size in frame, or the base rolls, creeps, or turns at any point after recording starts.
- **SOP rule broken:** Steps 1 to 10, the base is parked and locked before recording and stays still for the whole episode.
- **Coaching note:** park it, lock it, push-test it, then start recording. A base that drifts is a lost episode.

Violation: Approached from above or fouled the cabinet above

- **Visible cue:** a gripper comes down into the cavity from above instead of entering level from the front, or a wrist or forearm knocks, scrapes, or rests on the cabinet above or the top of

the door mouth on the way in or out.

- **SOP rule broken:** Steps 4, 5, and 7, every action at the microwave is a front approach, level, in and out through the door mouth.
- **Coaching note:** straight in from the front, straight out the same way. There is no way in from the top.

Violation: Leaned on or pushed the microwave

- **Visible cue:** a gripper, wrist, or forearm rests on the door, the door mouth, or the control panel; the microwave rocks or shifts; or a press on the door or a key moves the appliance.
- **SOP rule broken:** Steps 3 to 7, the microwave carries no weight and is never pushed out of place.
- **Coaching note:** the arm holds itself up. Press only as hard as the latch or the key needs.

Violation: Food plated wrong

- **Visible cue:** the **right gripper** puts fewer or more than three scoops on the plate, lifts heaped scoops instead of level ones, spreads the food into more than one mound, or leaves food on the plate rim or the counter.
- **SOP rule broken:** Step 1.3, put three level scoops on the plate's inner circle, laid one on another as one low mound.
- **Coaching note:** three level scoops, all in the middle, all in one mound. Food on the rim is food the grippers have to grip through.

Violation: Spoon handled or parked wrong

- **Visible cue:** the **right gripper** closes on the spoon bowl instead of the handle, drags the spoon across the counter, leaves it in the food tub, leaves it on the plate, or lays it down bowl down or away from where it started.
- **SOP rule broken:** Steps 1.2, 1.4, 9.1, and 9.3, take the spoon by its handle and lay it back flat on the counter where it started, bowl up, handle toward the front.
- **Coaching note:** handle to take it, same spot to put it back, bowl up both times.

Violation: Cover put on wrong

- **Visible cue:** the **right gripper** closes on the cover dome, rim, or a vent instead of the cover tab; the cover is dragged across the plate or tipped on at an angle; or it ends up with a gap under its rim, rocking when the gripper comes off, or sitting over the outer edge of the plate's rim.
- **SOP rule broken:** Step 2.2, take the cover by its tab, lower it straight down, and leave it seated on the plate's rim all round with the outer edge of the rim clear.
- **Coaching note:** tab only, straight down, then watch it settle before you let go. The outer rim is where the grippers go.

Violation: Door opened wrong or not to its stop

- **Visible cue:** the **left gripper** opens the door; the **right gripper** takes the door somewhere other than its handle; the door is jerked or swung fast enough to bang the cabinet run; the door is held by an arm instead of standing at its stop; or it is left part way open so the mouth is not clear.
- **SOP rule broken:** Steps 3.1 and 7.1, the **right gripper** takes the handle and swings the door round to its stop in one smooth move, and the door stands there on its own.
- **Coaching note:** right gripper, handle, smooth swing, all the way to the stop, then let go.

Violation: Plate taken with the wrong grip

- **Visible cue:** one gripper alone lifts the plate, a gripper closes on the cover, the inner circle, or the food instead of the outer rim, the two grippers close at different times, or they take the rim at an angle to each other instead of straight across.
- **SOP rule broken:** Steps 4.1 and 7.2, both grippers close on the plate's outer rim at the same time, one on each side, straight across from each other.
- **Coaching note:** both hands, opposite sides, rim only, closing together. A one-gripper plate tips.

Violation: Plate tilted, swung, or moved fast

- **Visible cue:** the plate tips out of level during a carry, swings between the grippers, turns between them, is snatched out of the cavity, or the cover shifts or the food slides while it is carried, going in or coming out.
- **SOP rule broken:** Steps 4.2 and 7.3, carry the plate level and straight, lifting only high enough to clear the surface and drawing it straight out of the door mouth slowly.
- **Coaching note:** level, low, and slow, both ways. A hot plate that tips is a spill you cannot undo.

Violation: Plate left off center on the turntable

- **Visible cue:** the plate sits nearer one cavity wall than the other, touches a wall, hangs over the edge of the turntable, or sticks out past the door mouth when the grippers come off.
- **SOP rule broken:** Step 4.3, nudge the plate until the gap to the cavity wall looks the same on the left as on the right, with its front edge inside the door mouth.
- **Coaching note:** an off-center plate catches the wall as soon as the turntable turns. Nudge it to the middle before you come out.

Violation: Door not shut and latched before the cycle

- **Visible cue:** the door springs back after the **right gripper** comes off, stands ajar or proud of the front, is swung shut fast enough to slam, or a key is pressed while the door still stands open or unlatched.

- **SOP rule broken:** Step 5.1, swing the door shut and press its front face flat and straight back at the handle side until it is latched, before any key is pressed.
- **Coaching note:** flat face, straight back, then watch it before you move away. A door left ajar will not run.

Violation: Cycle set or started wrong

- **Visible cue:** the **right gripper** presses keys other than **3** then **0**, presses two keys at once, rests on the panel between presses, presses **Start** before the time is set, or the **left gripper** presses any key.
- **SOP rule broken:** Steps 5.2 and 6.1, press **3** then **0** one key at a time with the **right gripper**, then press **Start**.
- **Coaching note:** time first, start second. One key at a time, straight in, straight back off, right gripper only.

Violation: Arms not clear or something touched while the cycle ran

- **Visible cue:** an arm stays over the counter, the control panel, or the door mouth while the cycle runs, an arm moves during the cycle, or a gripper touches the door, the panel, the food tub, the spoon, or the cover before the cycle has ended.
- **SOP rule broken:** Step 6.2, draw both arms back clear of the front and hold them still, touching nothing until the cycle has ended.
- **Coaching note:** hands off while it runs. The cycle is part of the episode, not a break in it.

Violation: Door opened before the cycle ended or before the stand

- **Visible cue:** the **right gripper** takes the door handle while the display is still counting down, the light is still on, or the turntable is still turning, or it opens the door straight after the beep without the 5 second stand.
- **SOP rule broken:** Step 7.1, wait for the beep, the display at zero, the light out, and the turntable stopped, then leave it alone for 5 seconds before the door is opened.
- **Coaching note:** let it beep, then count five. Opening early throws steam at the arms and stops the cycle short.

Violation: Hot part touched after the cycle

- **Visible cue:** after the cycle, a gripper touches the food, the cover dome or rim, the turntable, or the inside of the cavity, or an arm reaches into the cavity for anything other than the plate's rim.
- **SOP rule broken:** Steps 7.1 to 9.3, after the cycle the grippers touch only the plate rim, the cover tab, and the spoon handle.
- **Coaching note:** three things only. If it was in the microwave, it is hot.

Violation: Hot plate set down wrong

- **Visible cue:** the plate is dropped the last part of the way onto the counter, set down on the counter's front edge or away from where it started, set down on top of the spoon or the cover, or let go before it sits flat, or it rocks or slides after the grippers come off.
- **SOP rule broken:** Step 7.3, lower the plate straight down onto the spot it started from, set it flat, and only then open both grippers.
- **Coaching note:** all the way down, flat, then let go. Its own spot, every time.

Violation: Cover lifted or set down wrong

- **Visible cue:** the **right gripper** lifts the cover before it clears the plate rim, drags it across the plate, knocks the plate or the food with its rim, tips it so the steam goes toward the front edge, or sets it down dome down, somewhere other than where it started, or drops it onto the counter.
- **SOP rule broken:** Steps 8.2 and 8.3, lift the cover straight up by its tab until it clears the plate rim, tilt its far edge down and away from the front, and set it back on the counter dome up where it started.
- **Coaching note:** up first, then tip it away from you, then straight back to its own spot.

Violation: Plate not held still while it is worked

- **Visible cue:** the **left gripper** is not on the plate's rim while the cover goes on, while the cover comes off, or while the food is stirred, comes off it early, or the plate slides, turns, or rocks on the counter during any of them.
- **SOP rule broken:** Steps 2.1, 8.1, and 9.3, the **left gripper** holds the plate by its rim before the cover is lowered, and from before the cover is lifted until after the spoon is put back.
- **Coaching note:** hold it first, then work it. A plate that slides under a stir ends up on the floor.

Violation: Stirred wrong

- **Visible cue:** the **right gripper** turns the spoon fewer or more than three full turns, lifts the spoon out between turns, keeps the bowl above the plate instead of down against it, pushes food over the plate rim, or leaves the mound uneven.
- **SOP rule broken:** Step 9.2, lower the spoon to the plate and turn it three full turns with the bowl down, keeping the food inside the inner circle.
- **Coaching note:** bowl down, three turns, no lifting in between. The point is to move the edges into the middle.

Violation: Wrong order of work

- **Visible cue:** the cover goes on before the food, the door is opened before the plate is covered, the plate goes in before the door is open, the time is set before the door is latched, the plate is uncovered before it is out of the cavity, or the food is stirred before the cover is off.

- **SOP rule broken:** Steps 1 to 9, plate the food, cover it, open the door, center the plate, set the time, run the cycle, retrieve the plate, uncover it, stir it.
- **Coaching note:** the order is the task. Every step leaves the station ready for the next one.

Violation: Required check not followed

- **Visible cue:** a check named in a step is skipped, or a check is made and the fault it finds is left uncorrected — food on the rim, a cover not seated, a plate off center, a door that springs back, a wrong display reading, a rocking plate, or an uneven mound after the stir.
- **SOP rule broken:** Steps 1.1 to 9.3, run each check and correct what it finds by the retry written in that step.
- **Coaching note:** a check is not done until what it found has been put right.

Violation: Dropped, spilled, or knocked over

- **Visible cue:** the plate, the cover, the spoon, or food is dropped on the counter, in the cavity, or on the floor; the food tub is knocked out of place or tipped; or the plate is knocked off the turntable.
- **SOP rule broken:** Steps 1 to 9, nothing is dropped, spilled, or knocked out of its place, and every gripper leaves the door mouth straight out to the front.
- **Coaching note:** check the path and the landing place before the arm moves, and come out the way you went in.

Violation: Wrong arm used

- **Visible cue:** the **left gripper** takes the spoon, the cover, the door handle, or a key on the control panel; the **right gripper** holds the food tub or the plate still while the other works; the plate is taken by one gripper alone; or anything is handed from one gripper to the other.
- **SOP rule broken:** Steps 1 to 9, the **right gripper** works the spoon, the cover, the door, the panel, and the stir, the **left gripper** holds the tub and the plate, and both take the rim together for the two carries.
- **Coaching note:** right arm works, left arm holds, both arms carry the plate, and nothing changes hands.

Violation: Wrong episode ending

- **Visible cue:** the episode ends with the plate still in the cavity, the plate still covered, the food unstirred, the spoon or the cover left out of place, a spill left on the counter, an arm short of home, or a gripper not fully open.
- **SOP rule broken:** Step 10, confirm the plate, the microwave, the counter, the spills, and the base, then return both arms home with grippers open and stop recording.
- **Coaching note:** confirm first. Homing is the last thing the arms do.

Failures that are not violations

These failures are not caused by how the task was run. Log them as system issues, discard the episode, and never use them for coaching.

- **Recording stopped or paused during the episode** (recording system).
- **Camera dropped frames or lost its feed** (capture system).
- **Camera cannot read the turntable, the plate, or the control panel**, so whether the plate was centered, how the food was plated, or which keys were pressed cannot be judged.
- **Hardware fault on an arm**: gripper failure, drift, controller caused collision, or motor error.
- **Heat pad fault**: a pad that comes loose, slips on the gripper finger, or will not hold a correctly gripped plate rim.
- **Base fault**: a brake or lock that will not hold, or a base that rolls, creeps, or turns with the lock set.
- **Faulty microwave**: one that will not start with a correct key sequence, that stops mid-cycle, that does not beep or clear its display at the end, or whose light or turntable does not stop when the door is opened.
- **Faulty door**: a door whose latch will not hold, that will not stand open at its stop, or that swings shut on its own.
- **Faulty turntable**: one that is not seated, that will not turn, or that turns unevenly enough to walk a correctly centered plate into a wall.
- **Door mouth too tight**: a mouth so low or so narrow that both grippers cannot enter level from the front holding the plate, or cannot reach the back of the turntable.
- **Faulty cover**: one whose tab comes off in a correct grip, that melts, warps, or sags in the cycle, or whose vents are blocked so it lifts off the plate while the cycle runs.
- **Defective plate**: one that rocks on the counter, that cracks or crazes in the cycle, that comes out too hot for a padded gripper to hold, or that slips out of a correct rim grip.
- **Food will not behave**: food too runny to hold a mound, too stiff to lift on a level scoop, or that boils over inside the cover in a correct cycle.
- **Food tub will not stay put**: a tub that slides or tips on the counter when the spoon is drawn through it.
- **A place turns out to sit outside its arm's comfortable reach** with the base correctly parked, so the food tub, the spoon, the cover, the plate's rim, the door handle, a key on the control panel, or the back of the turntable cannot be reached without extending or folding the arm.

Annotation subtasks (from SOP)

1. Hold the food tub still
2. Take the spoon off the counter

3. Put one level scoop of food on the plate
4. Lay the spoon back on the counter
5. Take hold of the plate's rim to hold it still
6. Lift the cover by its tab
7. Set the cover down over the plate
8. Swing the microwave door open to its stop
9. Take the plate's rim with both grippers
10. Carry the plate into the cavity and set it on the turntable
11. Nudge the plate centered on the turntable
12. Swing the door shut and press it home
13. Press one key on the control panel
14. Stand clear while the cycle runs
15. Carry the plate out of the cavity onto the counter
16. Lift the cover off the plate
17. Set the cover back on the counter
18. Stir the food three full turns
19. Return both arms home and end the episode